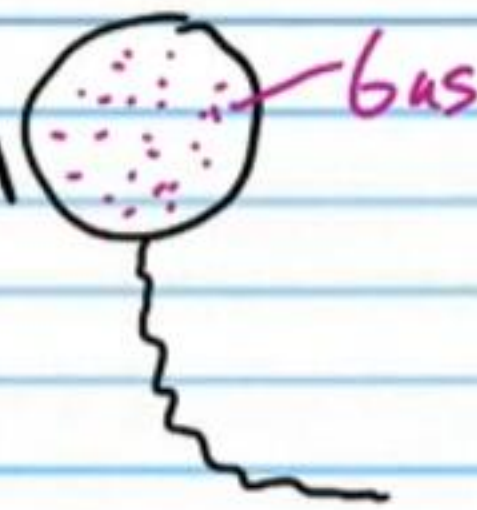


# Applications of Algebra (Part I) Homework

① The volume of the gas in the balloon at the right is directly proportional to the temperature, such that  $V = 7.45T$ , where  $V$  is the volume in milliliters and  $T$  is the temperature in Kelvins.



a) If the temperature of the gas in the balloon is 279 kelvins, what is the volume of the gas?

b) What is the proportionality constant?

c) If the volume of the gas in the balloon is 1,800 milliliters, what is the temperature (assuming that atmospheric pressure is constant)?

② The time that it takes to complete a job is inversely proportional to the number of people that participate, such that  $t = 15/p$ , where  $t$  is the number of days required to finish the job and  $P$  is the number of people working.

a) If 3 people attempt to finish the job, how many days will it take?

b) What is the proportionality constant?

c) If it takes 15 days to finish the job, how many people participated?

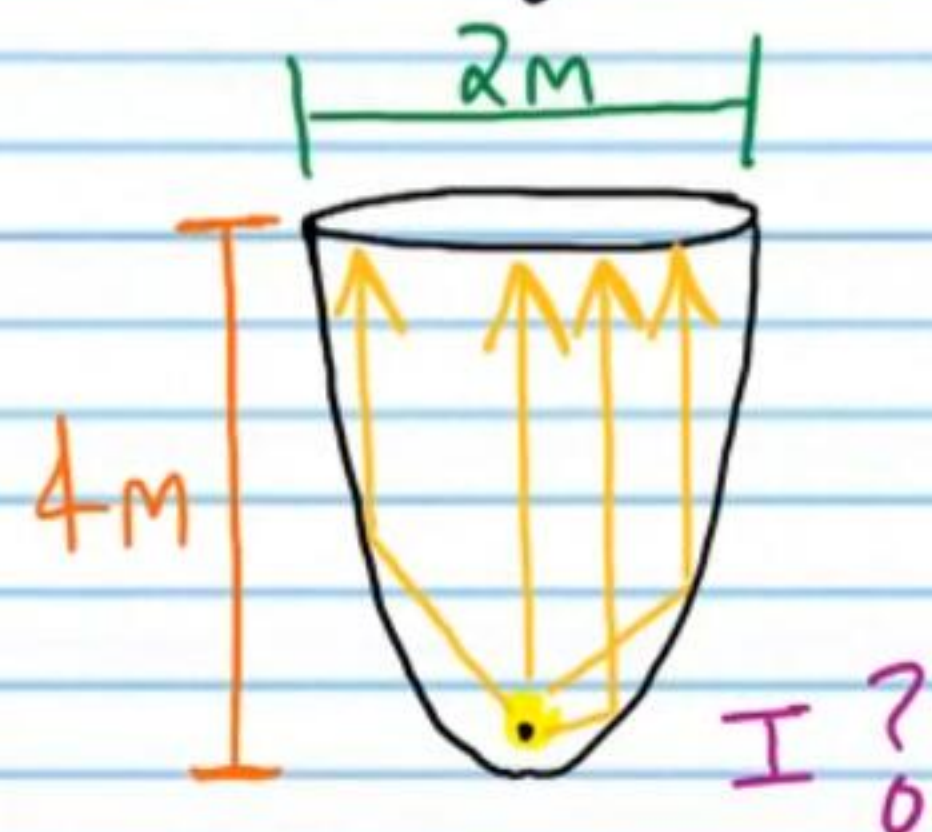
③ If  $y$  is directly proportional to  $x$ , and  $x=3$  when  $y=24$ , find  $x$  when  $y=98$ .

④ The cost of a gem (in dollars) is inversely proportional to the number of gems that are available. If a particular gem costs \$82 per ounce when there are 8,000 on the market, what would the cost be per ounce if there were 20,000 on the market?

⑤ Rhonda has quarters, dimes, and nickels in her purse. If there are 39 coins worth a total of \$5.65, and there are 4 more dimes than twice the number of nickels, how many of each type of coin are there?

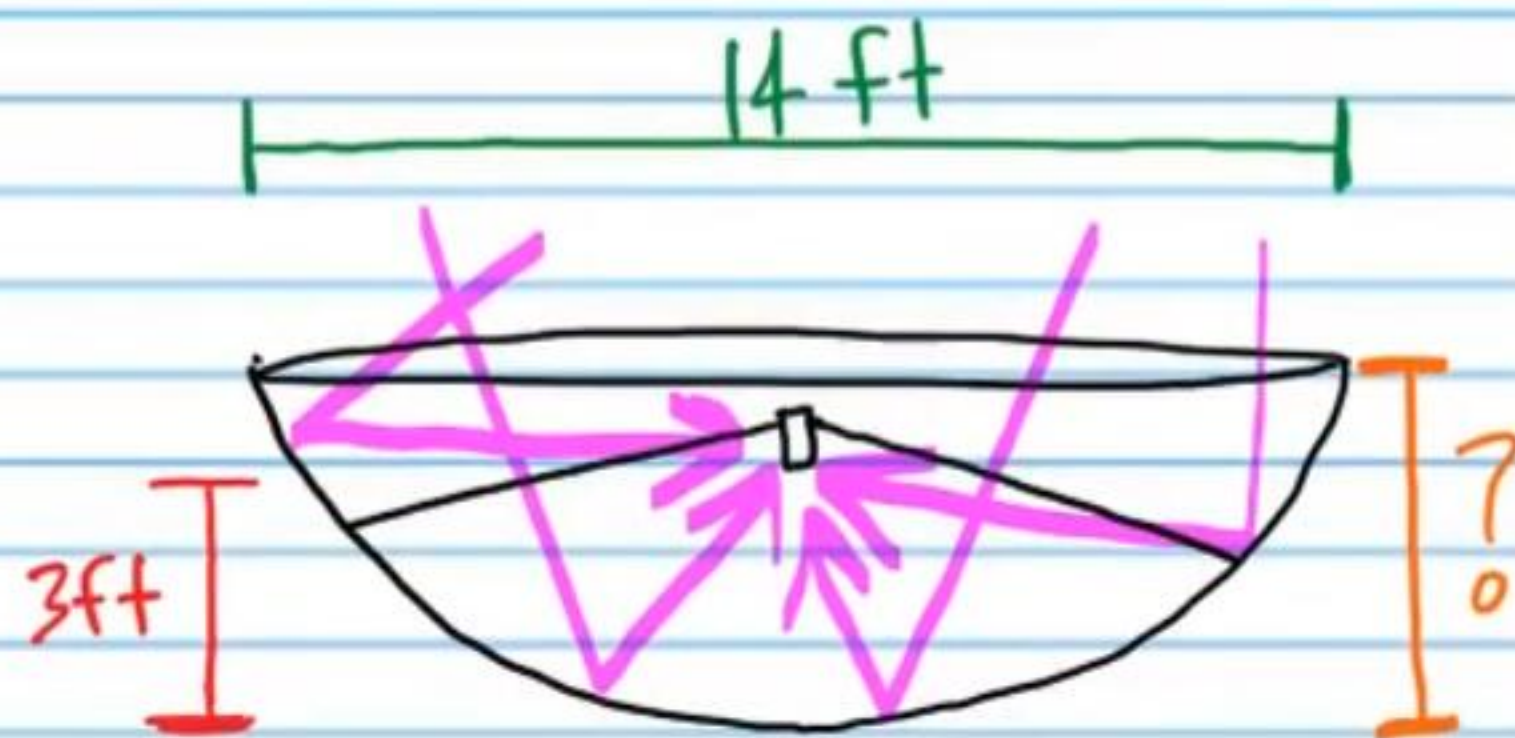
⑥ While scuba diving, Marcus found 58 coins worth \$3.70 total. If he found only quarters, nickels, and dimes, and there were 4 times as many dimes as quarters, how many of each type of coin did he find?

⑦ A searchlight is shaped like a paraboloid. If the opening is 2 meters wide and the depth of the recess is 4 meters, where should the light be placed so that the rays are reflected in the right direction?





⑧ A satellite dish is shaped like a paraboloid. Signals from a satellite hit the dish and are reflected to a single point where a receiver is located. If the receiver is 3 feet from the vertex of the dish, and the opening of the dish is 14 feet wide, how long (i.e. deep) should the dish be?



## Answers

① a)  $2,078.55$  milliliters

b)  $7.45$

c)  $241.6$  Kelvins

② a)  $5$  days

b)  $15$

c)  $1$  person

③  $x = 12.25$

④  $\$32.8/\text{oz}$

⑤  $18$  dimes,  $14$  quarters, and  $7$  nickels

⑥  $8$  dimes,  $2$  quarters, and  $48$  nickels

⑦  $\frac{1}{16}$  of a meter from the vertex of the paraboloid.

⑧  $4\frac{1}{12}$  feet long